

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
A-LEVEL · FURTHER MATHEMATICS

FURTHER MATHEMATICS 9187/1

PAPER 1 November 2023

3 hours

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2023 9187/1.

Additional materials: A calculator may be used unless a question forbids it.

INSTRUCTIONS: Answer all questions. Full marks are not given for answers only — show method.

- 1.** $z = 3 + 4i$. Find $|z|$, $\arg z$, and $\bar{z}$. Express $1/z$ in the form $a+bi$. **[10]**

- 2.** $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$. Find $\det A$, A^{-1} , and A^2 . **[10]**

- 3.** Prove by induction that $1 + 2 + \dots + n = n(n+1)/2$ for $n \geq 1$. **[10]**

- 4.** Write $1 + i$ in modulus-argument form and find $(1+i)^4$ using De Moivre. **[8]**

- 5.** Use identities to simplify $\sin 3\theta$ in terms of $\sin \theta$ (or state $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$). **[8]**

- 6.** Show that $(i+j) \times (i+2j+k) = i - j + k$ or compute the cross product of $a=(1,1,0)$, $b=(1,2,1)$. **[8]**

- 7.** Standard Further Pure question on Maclaurin series. Show full working. **[8]**

- 8.** Standard Further Pure question on Reduction formula sketch. Show full working. **[8]**

- 9.** Standard Further Pure question on Eigenvalues of $(2,1;1,2)$. Show full working. **[8]**

- 10.** Standard Further Pure question on Second order DE $y''+y=0$. Show full working. **[8]**

- 11.** Standard Further Pure question on Further partial fractions. Show full working. **[8]**

12. Standard Further Pure question on Argand diagram loci. Show full working. **[8]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).